PHIL1002: Introduction to Critical Thinking

Lecture 3A: Argument Structure–II

1

Agenda: Argument Structure

Part 1
1. The basics of standardising arguments
2. Tips and tricks

Part 2
1. Missing Premises
2. Revised Standard Form

2

Missing premises

This is the problem of what you should do when you find logical gaps in an argument.

Conclusions are not the only things which people omit when they are reasoning. They miss out premises too.

Often these aren't mentioned because they're simply not worth mentioning; they may just be too obvious to need pointing out.

But sometimes they can be very controversial.

3

Example...

The Government should privatise the Water Corporation, because the state needs an industry which can act quickly and flexibly in response to climate change.

Missing premises??

4

Example…

The likely omitted premises would be something like these:

1. State-controlled industries cannot act quickly and flexibly

2. The Water Corporation is a state-controlled industry

3. The State Government should respond to the state's needs

5

Example…

1'. State-controlled industries actively prevent climate change initiatives

6

The Principle of Charity

When interpreting someone else's argument, always interpret it in such a way as to give them the strongest possible reasoning compatible with the content of what they have actually said.

7

The all-purpose missing premise...

8

Example…

1. In capitalist economies the means of production are privately owned

So

2. Capitalist states are inherently anti-democratic.

9

Example…

If in capitalist economies the means of production are privately owned, then capitalist states are inherently anti-democratic

10

Example…

1. *In capitalist economies the means of production are privately owned*

[2. If in capitalist economies the means of production are privately owned, then capitalist states are inherently anti-democratic]

So

3. Capitalist states are inherently anti-democratic

11

Example…

1. The astronauts brought back some cheese from the moon
So
2. The moon is made of cheese

And

1. China, Japan and Russia form a significant part of Australia's export markets
So
2. It is important to the Australian economy that close attention be paid to Asian economic developments

12

No perpetual motion machine!

Putting an argument into standard form, together with the supply of missing premises, is a preliminary to the evaluation of that argument.

The question to ask about the addition of the all-purpose premise is, what does it do to **show** that the argument is a good one, or to **improve** it and make it a good one?

And the answer to this question is that it does **nothing.**

13

Logical gap filler?

A, therefore B

should you add a missing premise to make it read like this:

A; if A, then B; therefore, B

Has this new version sealed a gap in the earlier one?

NO IT HASN'T

Why not?

14

Example…

1. *The astronauts brought back some cheese from the moon*
2. *If astronauts brought back some cheese from the moon, then the moon is made of cheese*

So

3. The moon is made of cheese

15

Begging the question…

For the same reason, you can't do an argument any good by adding to it the premise:

This argument proves its conclusion

The added premise simply claims, without giving further supporting reasons, that the argument is a good one. But that was the question that was in doubt in the first place.

This is your first introduction to the **fallacy of begging the question.**

16

More on missing premises

There's another way in which one might be tempted to start supplying unnecessary missing premises.

The end of a thing is its goal.
Death is the end of life.
So,
Death is the goal of life.

Any argument is going to have to rely on the meanings of the words in which it is expressed.

The meanings of the crucial words must stay the same throughout so that you don't end up with arguments like this one:

17

The moral of the story...

…be **very** sparing about adding missing premises–when in doubt, leave it out.

18

Agenda: Argument Structure

Part 1
1. The basics of standardising arguments
2. Tips and tricks

Part 2
1. Missing Premises
2. Revised Standard Form

19

Govier's standardisation method...

Is limited for a couple of reasons.

First, it can obviously be quite laborious.

The second limitation…requires a longer story.

20

Example…

(e) *Chimps are susceptible to all the infectious diseases of man*

So

(f) *Wherever the populations of chimps are near new human settlements, these chimps are endangered by epidemics*

(c) *Forests are cleared to make way for cultivation*

(d) *Food trees are poisoned to leave space for better timber trees*

So

(b) *With the spread of agriculture and forestry, the habitat as well as the lives of chimpanzees are threatened*

So

(a) *There is a shadow that is spreading over the chimpanzee today*

21

Example…

(e) *Chimps are susceptible to all the infectious diseases of man* **[prem.]**

So

(f) *Wherever the populations of chimps are near new human settlements, these chimps are endangered by epidemics* **[e]**

(c) *Forests are cleared to make way for cultivation* **[prem.]**

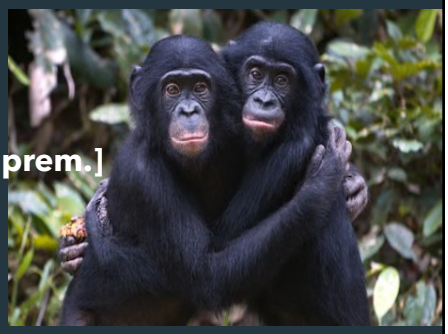
(d) *Food trees are poisoned to leave space for better timber trees* **[prem.]**

So

(b) *With the spread of agriculture and forestry, the habitat as well as the lives of chimpanzees are threatened* **[c, d, f]**

So

(a) *There is a shadow that is spreading over the chimpanzee today* **[b]**



22

Revised Standard Form…

…gives us a lot more information about the argument than ordinary standard form does.

And it also gives us a lot more freedom in how we set out the argument.

We can, for instance, list **all** the premises at the start of the argument: each conclusion doesn't have to be introduced **immediately** below what it's been inferred from.

Standard form relies on spatial adjacency to show the relationships of the various steps of the argument. **Revised standard form** removes that restriction.

23

Example…

(e) *Chimps are susceptible to all the infectious diseases of man* **[prem.]**

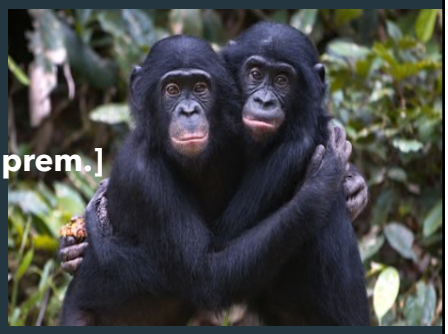
(f) *Wherever the populations of chimps are near new human settlements, these chimps are endangered by epidemics* **[e]**

(c) *Forests are cleared to make way for cultivation* **[prem.]**

(d) *Food trees are poisoned to leave space for better timber trees* **[prem.]**

(b) *With the spread of agriculture and forestry, the habitat as well as the lives of chimpanzees are threatened* **[c, d, f]**

(a) *There is a shadow that is spreading over the chimpanzee today* **[b]**



24

Convergent or linked?

Two ways in which a statement can be derived from two other statements.

CONVERGENT: a conclusion is derived from each premise of the argument quite independently of its derivation from the other.

LINKED: a conclusion is derived only by taking the two statements together, each of them on its own giving no independent support to the conclusion but giving some support when combined.

25

Convergent or linked?

CONVERGENT STRUCTURE

Premise Premise

↘ ↙

Conclusion

LINKED STRUCTURE

Premise + Premise

↓

Conclusion

26

Example…

(e) *Chimps are susceptible to all the infectious diseases of man* **[prem.]**

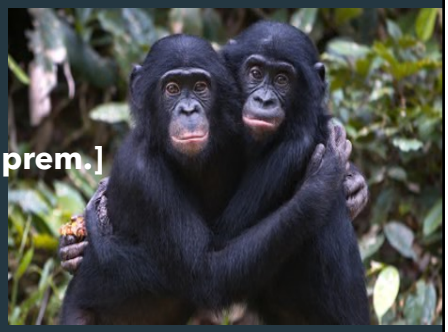
(f) *Wherever the populations of chimps are near new human settlements, these chimps are endangered by epidemics* **[e]**

(c) *Forests are cleared to make way for cultivation* **[prem.]**

(d) *Food trees are poisoned to leave space for better timber trees* **[prem.]**

(b) *With the spread of agriculture and forestry, the habitat as well as the lives of chimpanzees are threatened* **[c, d, f] LINKED]**

(a) *There is a shadow that is spreading over the chimpanzee today* **[b]**



27

Argument microstructure….

All devils ride dragons

All dragons are fire-breathers

Therefore

All devils ride fire-breathers

28

Next lecture

29